MetaboGuard Progress Report

Self-supervised metabolic phenotyping, longitudinal simulation,

and resolution of the biomarker clustering abstention

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Executive conclusion. Since the 4 August meeting, MetaboGuard has moved from a cross-sectional proof of concept to a reproducible research platform with fail-closed data checks, label-free representation learning, evidence-linked biomarker interpretation, exploratory clustering, a simulation-only longitudinal risk pipeline, and a password-protected professor dashboard. The principal negative result is scientifically useful: no candidate clustering passed all quality, stability, outlier and nuisance-control gates. The dominant recovered structure was survey cycle, missingness and assay availability rather than a defensible metabolic phenotype. The correct next step is not to relax the gates. It is to harmonise assays across cycles, test clustering within cycles against correlation-preserving null data, transport frozen clusters to held-out cycles, and adopt continuous latent factors if discrete clusters remain unsupported.

Clinical boundary. MetaboGuard remains non-diagnostic. NHANES is cross-sectional, TCGA is post-diagnosis, and the longitudinal future-risk pipeline is trained only on synthetic Synthea records. No output currently estimates a real patient's risk of developing cancer or diabetes.

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1 Project objective and evaluation posture

MetaboGuard investigates whether combinations of routine metabolic, biochemical and behavioural measurements contain reproducible patterns that could support clinician-reviewed early-warning research. The project deliberately separates four questions:

1. whether the input data are reliable and comparable;
2. whether a label-free model learns a useful metabolic representation;
3. whether discrete, reproducible phenotypes exist;
4. whether a future longitudinal cohort can support calibrated event risk.

The present report covers work completed since the 4 August 2026 professor meeting. It uses committed repository records, frozen JSON reports and project documentation. The current clustering report is tied to dataset fingerprint `af790f9f...d6b8b7c`; the deployed snapshot clusters a 16-dimensional frozen latent representation using persisted patient-grouped splits.

2 Changes completed since the last meeting

2.1 Delivery timeline

Table 1: Major committed milestones after the 4 August meeting.

Date	Change delivered
4 Aug	Fail-closed cancer coding and leakage tests; self-supervised training tests; benchmark guide; model card; reproducibility guide.
4 Aug	Exploratory clustering, reliability reporting, evidence claims contract, and documented professor feedback.
4–5 Aug	Synthetic patient generation, protected professor dashboard, serverless NumPy inference, deployment security tests, Vercel configuration and Git deployment assets.
5 Aug	Versioned longitudinal schema, Synthea generation pipeline, future-risk protocol, deterministic synthetic histories and simulation-only dashboard panel.
8 Aug	Portable future-risk export; enhanced probe interpretation; refactoring; offline PyTorch retraining with NumPy serving compatibility.

2.2 System architecture

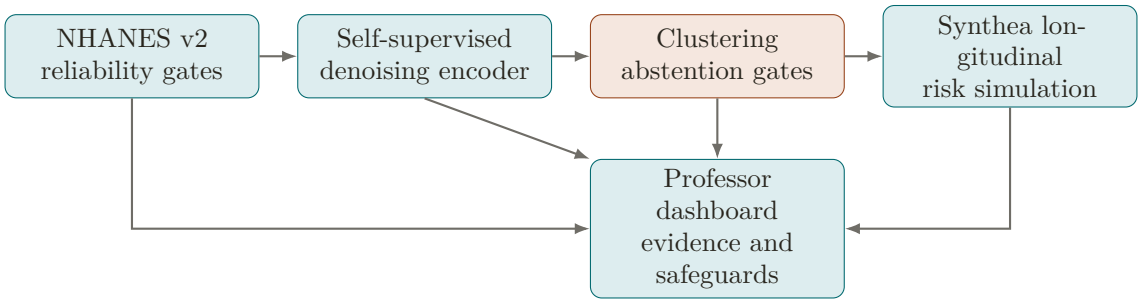


Figure 1: Current research architecture. Orange denotes a deliberate abstention boundary rather than a failed software path.

2.3 Data integrity and reliability

The reliability pipeline now records provenance, schema, plausible ranges, duplicate handling, feature coverage, missingness by split and cycle, assay-cycle drift, label confidence, leakage, survey-weight applicability and capability state. The current report passes hard validation and places features into four explicit eligibility tiers.

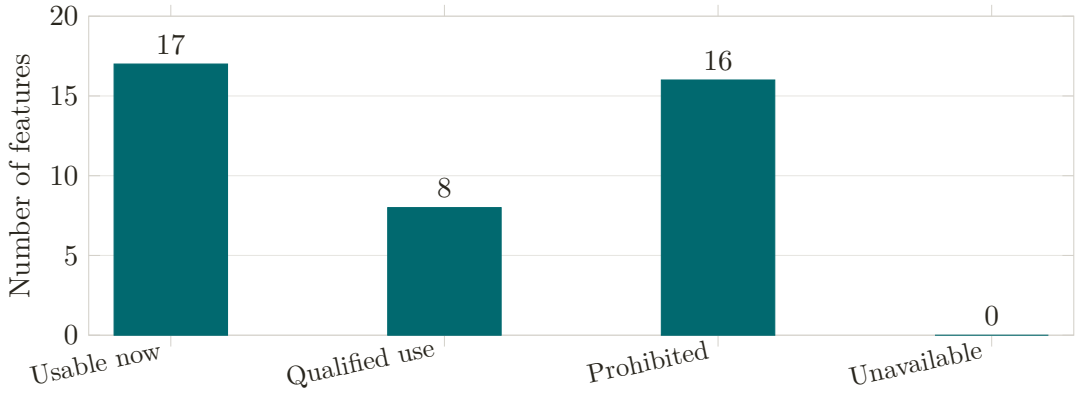


Figure 2: Feature eligibility after reliability checks. Prohibited variables are outcomes, label-derived variables or post-diagnosis context and are blocked from the encoder.

2.4 Self-supervised representation and benchmarking

The core model remains a label-free denoising encoder. Preprocessing is fitted on the training partition only; split IDs, seeds, fingerprints and model metadata are persisted. The deployment uses NumPy-exported weights, while training uses PyTorch. PCA and Isolation Forest were added as transparent baselines. This separation demonstrates that dependency-light serving is not equivalent to training a NumPy-only model.

The professor dashboard adds protected access, synthetic-patient probing, de-identified CSV validation, reliability and cluster reports, source-linked evidence, methodology explanations and simulation-only future-risk views. No uploaded file is retained.

2.5 Biomarker evidence and claims contract

A machine-readable evidence catalogue and claims contract now distinguish:

- patient observations;
- model associations;
- published biomarker evidence;
- causal claims, which remain unestablished unless supported by an appropriate causal design.

This prevents diagnostic case-control biomarker performance from being misrepresented as pre-diagnostic performance. It also prevents deviation scores, cluster assignments and synthetic future-risk values from being described as clinical probabilities.

2.6 Longitudinal schema and future-risk simulation

The new longitudinal schema represents timestamped observations, visits, index dates, outcomes, censoring, provenance and missingness. The synthetic cohort uses official Synthea records, which

are suitable for software verification but not clinical validation (Walonoski et al., 2018; Chen et al., 2019).

The committed pooled cohort contains 1,542,237 event rows and 5,784 eligible patients, with a median of 10 visits per patient. It supports only horizons that clear the pre-specified 50-event and 50-non-event gate.

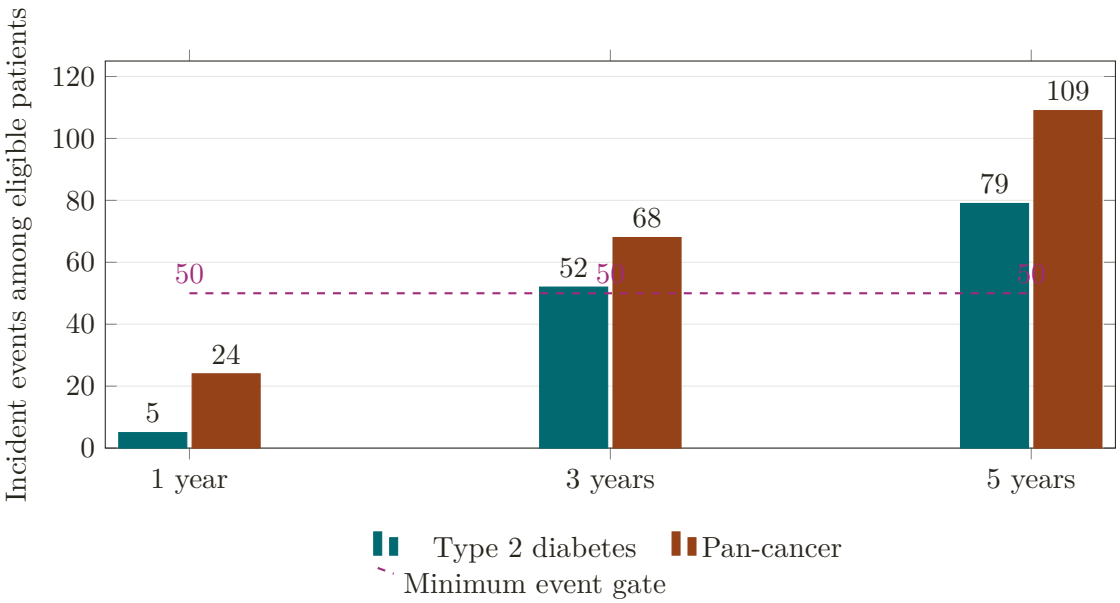


Figure 3: Synthea event counts. One-year horizons abstain; eligible horizons are simulation-only and all held-out event counts remain small.

The risk pipeline includes horizon logistic regression, cause-specific discrete-time hazards, gradient-boosted trees and an experimental temporal GRU. Temporal models must lose at least 0.02 AUROC when visit order is reversed. Type 2 diabetes GRUs failed this gate; pan-cancer GRUs passed at three and five years. Every metric is explicitly marked synthetic and underpowered, and the clinical endpoint remains HTTP 409.

3 Primary finding: no defensible biomarker clusters

3.1 What the current run found

The deployed clustering report uses 4,000 training rows, five bootstrap rounds and K-means with $K = 2, 3, 4$. All candidates fail at least one gate.

Table 2: Exact clustering diagnostics from the committed all-adults report.

Candidate	Silhouette	Bootstrap ARI	Min. Jaccard	Cycle V	Main failures
K-means $K = 2$	0.492	0.238	0.305	0.482	Cycle dominated; unstable; one cluster dissolves.
K-means $K = 3$	0.389	0.450	0.414	0.281	Tiny cluster; missingness dominated; outlier sensitive.
K-means $K = 4$	0.253	0.965	0.867	0.495	Tiny cluster; cycle, missingness and assay dominated; outlier sensitive.

The result is `no_stable_clusters`. This is not a contradiction: $K = 4$ has strong geometric stability but fails validity controls, while $K = 2$ has adequate geometric separation but fails resampling stability and cycle control. Stable clusters can still be stable artefacts (Hennig, 2007); consensus procedures can find crisp partitions in correlation-preserving null data (Senbabaoglu et al., 2014; John et al., 2020).

3.2 Why this happened

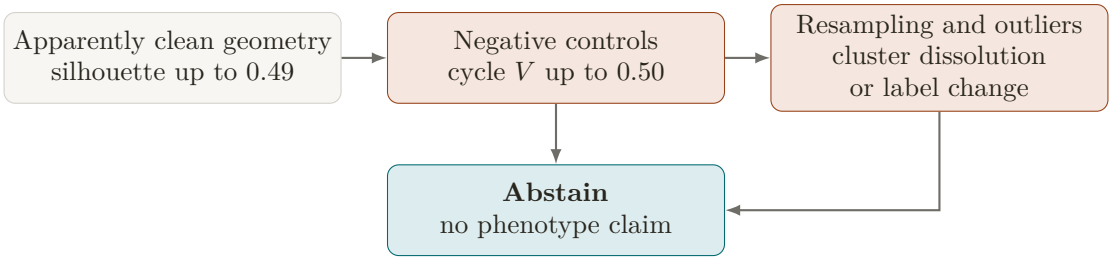


Figure 4: Why the abstention is a safety success. Geometry is evaluated before, but never allowed to override, nuisance and stability checks.

The most likely causes are:

1. **Assay and protocol changes across cycles.** NHANES documents instrument changes and publishes bridging equations for some biomarkers, including insulin (NCHS, 2011).
2. **Structural missingness and subsampling.** Fasting analytes and specialist assays are available only in subsamples and selected cycles.
3. **Pooling before harmonisation.** Cycle is a large, structured signal; official guidance requires method comparability and appropriate multi-cycle weights before combining data (NCHS, 2026a,c).
4. **Possibly continuous rather than discrete biology.** A metabolic gradient may be real even when hard clusters are not. Continuous clinical variables can outperform patient subgroup labels for prognosis and treatment selection (Dennis et al., 2019).
5. **Biomarker coverage.** Routine NHANES measurements do not include a consistent multi-omics panel or verified pre-diagnostic cancer outcomes.

Interpretation for the paper. The current result should be reported as: "No discrete metabolic phenotype passed the pre-specified validity gates in pooled, unharmonised multi-cycle NHANES." It should not be reported as: "metabolic phenotypes do not exist."

4 Plan to resolve the clustering issue

4.1 P0: establish whether the signal survives harmonisation

1. **Freeze an analyte-by-cycle provenance table.** Record instrument, method, limit of detection, age eligibility, fasting requirement, subsample weight and bridging equation for every feature. Use only features with stable methods or documented conversion equations (NCHS, 2026c).
2. **Create raw and harmonised matrices.** Apply CDC bridging equations first, then transformation and scaling. Compare pooled robust scaling with within-cycle robust scaling.

3. **Require nuisance reduction before clustering.** A cross-validated survey-cycle classifier should have AUC ≤ 0.60. Cycle association for a promoted cluster should be $V \leq 0.20$.
4. **Split cycles into discovery and held-out sets.** No held-out cycle may influence preprocessing, K selection or hyperparameters.

4.2 P0: test cluster existence against a matched null

Run K-means, Gaussian mixtures and fuzzy c-means within each discovery cycle for $K = 2, \ldots, 8$, with at least 25 seeds. Classical consensus clustering is not enough because it can partition cluster-less data (Senbabaoglu et al., 2014). Use M3C-style correlation-preserving null datasets and at least 100 Monte Carlo replicates (John et al., 2020).

Promote a discrete phenotype only when:

- Monte Carlo $p < 0.05$ at K, with adjacent $K - 1$ and $K + 1$ nulls also rejected;
- every cluster is at least 5% and contains at least 30 participants in every included cycle;
- bootstrap ARI, seed ARI and outlier/noise-replacement ARI are each at least 0.60;
- cluster-wise mean Jaccard is at least 0.75, raising the present threshold from "not dissolved" to "well recovered" (Hennig, 2007);
- cross-cycle transport ARI is at least 0.60;
- nuisance associations remain below pre-specified thresholds and their null distributions.

4.3 P0/P1: treat missingness and survey design explicitly

Use three parallel missing-data analyses:

1. complete-case clustering within common assay-availability strata;
2. multiple imputation with at least 20 completed datasets and a documented MAR/MNAR sensitivity analysis (Sterne et al., 2009; van Buuren and Groothuis-Oudshoorn, 2011);
3. a latent-variable model with native missing-modality handling.

Cluster discovery can remain unweighted as a declared exploratory analysis, but cluster prevalences and associations must use the NHANES weight, PSU and strata design. NCHS shows that unweighted estimates can differ materially from design-weighted estimates (NCHS, 2026a,b).

4.4 P1/P2: external and longitudinal validation

Freeze the selected centroids or posterior model and assign held-out cycles without refitting. A transported cluster must add predictive information beyond a continuous-feature comparator. This matters because externally replicated diabetes clusters have not consistently outperformed simple continuous predictors (Ahlqvist et al., 2018; Dennis et al., 2019).

If compatible clinical, metabolomic, proteomic and genetic blocks become available, use multi-view factor analysis first and similarity network fusion only on a defensible shared subsample (Argelaguet et al., 2018; Wang et al., 2014). A real longitudinal cohort remains necessary to test incident outcomes.

4.5 Pre-registered continuous fallback

If the null of one continuous population cannot be rejected after P0 harmonisation, the project should not force clusters. The fallback deliverable is a continuous metabolic factor or graded membership score. This is still aligned with early-warning research and may be more useful than hard subgroup labels (Dalmaijer et al., 2022; Dennis et al., 2019).

5 Eight-week delivery plan

Table 3: Recommended work programme to resolve the clustering question.

Week	Workstream	Main actions	Deliverable / gate
1	Provenance and design freeze	Analyte-cycle table; survey design; discovery/held-out cycles; seal outcomes; register all methods and gates.	Frozen analysis plan.
2	Harmonisation	CDC bridging equations; feature intersection; pooled and within-cycle standardisation; nuisance classifiers.	Cycle AUC ≤ 0.60.
3	Null calibration	Intercept-only ComBat sensitivity; 100 correlation-preserving null datasets; null distributions for silhouette, PAC, DB, CH and ARI.	Null reference table.
4	Missingness arms	Availability-stratified complete cases; 20+ imputations; native-missing factor model; compare assignments.	Across-imputation ARI and missingness verdict.
5	Within-cycle discovery	$K = 2 \ldots 8$, multiple algorithms, 25+ seeds, M3C tests, 50+ stability resamples and noise replacement.	First formal go/no-go.
6	Cross-cycle consensus	Transport centroids/posteriors between discovery cycles; continuous multi-view factors as fallback.	Cross-cycle ARI matrix.
7	Held-out transport	Assign untouched cycles without refitting; design-weighted cluster prevalence and uncertainty.	Transport report.
8	Outcome validation	Unseal outcomes once; compare clusters against continuous features; final decision and paper-ready negative result if needed.	Decision memo and manuscript update.

6 Risks, decisions and requests to the team

Decision requested	Recommendation
Primary clustering objective	Approve a formal cluster-existence test, not a target number of clusters.
Cycle window	Select cycles with the strongest assay comparability; hold later cycles out.
Survey statistics support	Assign biostatistical review for multi-cycle weights, PSU/strata inference and multiple imputation.
Cluster gate update	Raise promoted-cluster Jaccard to 0.75 and replace a fixed null-gain rule with a Monte Carlo p-value.
Fallback product	Pre-approve a continuous metabolic factor if discrete clusters remain unsupported.
External data	Prioritise access to a real longitudinal cohort and consistent multi-omics blocks.

7 Conclusion

The project has made substantial engineering and methodological progress: corrected data, leakage controls, reproducible self-supervised training, transparent baselines, evidence provenance, cautious clustering, a versioned longitudinal simulation pipeline and a professor-facing deployment. The clustering abstention is the most important finding because it prevents a plausible-looking but invalid phenotype claim.

The next phase should answer a narrower and publishable question: *after assay harmonisation, complex-survey handling and correlation-preserving null calibration, is there evidence for discrete, transportable metabolic phenotypes?* If the answer remains no, a continuous factor model is the scientifically correct output.

A Metric glossary

Metric	Interpretation
Silhouette	Separation relative to within-cluster compactness. Must be compared with a matched null; high values alone do not establish biological structure.
Adjusted Rand index	Chance-corrected agreement between two assignments. Used for seeds, bootstrap refits, outlier sensitivity and cross-cycle transport.
Cluster-wise Jaccard	Recovery of each individual cluster after resampling. Values ≤ 0.50 indicate dissolution; ≥ 0.75 is the proposed promotion gate.
Cramér's V	Association between cluster labels and categorical nuisance variables such as survey cycle.
Permutation/null gain	Difference between the observed cluster metric and a correlation-preserving null distribution.

B Safe statements for presentations and/or showcases

- "The system found candidate geometric partitions, but none passed all stability and nuisance-control gates."
- "The dominant recoverable structure was survey-cycle and assay availability, not a validated biological phenotype."
- "The abstention is a safety result, not a software failure."
- "Future-risk functionality has been engineered and tested on synthetic longitudinal records only; it is not a patient-risk model."
- "The next experiment will test cluster existence after harmonisation against a correlation-preserving null and held-out cycles."

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